



Cambridge (CIE) IGCSE Maths: Extended



Your notes

3D Pythagoras & Trigonometry

Contents

* 3D Pythagoras & Trigonometry

3D Pythagoras & Trigonometry



Your notes

3D Pythagoras & Trigonometry

How do I use Pythagoras' theorem in a 3D shape?

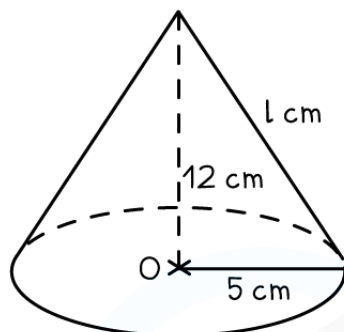
- You can often find **right-angled triangles** within 3D shapes
 - If two sides of the triangle are known, you can use **Pythagoras' theorem**



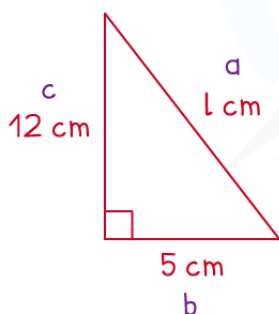
Your notes

PYTHAGORAS' THEOREM IN 3D

e.g. A CONE HAS BASE RADIUS 5 cm AND A PERPENDICULAR HEIGHT OF 12 cm.



FIND THE LENGTH, l cm, OF THE SLANTED HEIGHT OF THE CONE



DRAW THE 2D TRIANGLE OUT TO MAKE IT EASIER TO SEE.

$$a^2 = b^2 + c^2$$

$$l^2 = 5^2 + 12^2$$

$$l^2 = 25 + 144$$

$$l^2 = 169$$

$$l = 13 \text{ cm}$$

USE PYTHAGORAS' THEOREM TO SOLVE THE PROBLEM.

Copyright © Save My Exams. All Rights Reserved

Is there a 3D version of the Pythagoras' theorem formula?

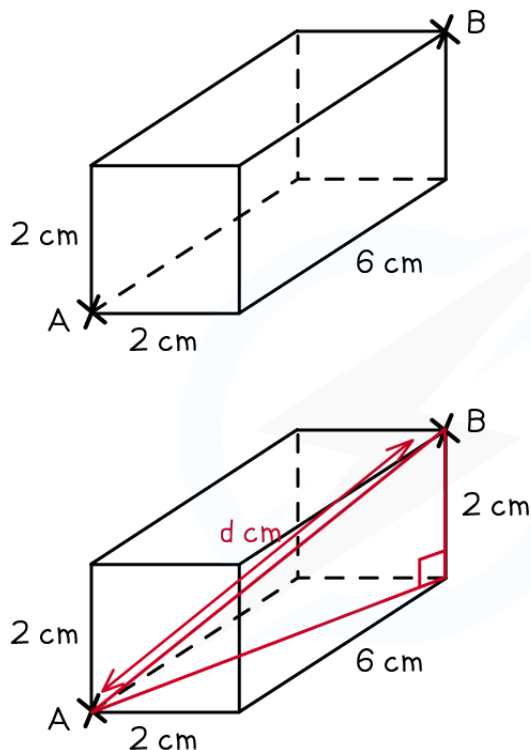
- There is a **3D version** of Pythagoras' theorem: $d^2 = x^2 + y^2 + z^2$



Your notes

- d is the distance between two points
- x , y and z are the distances in the **three different perpendicular directions** between the two points

e.g. FIND THE DISTANCE BETWEEN A AND B.



$$d^2 = 2^2 + 6^2 + 2^2 = 44$$

LENGTH OF AB IS $\sqrt{44} = 6.63$ (2 DP)

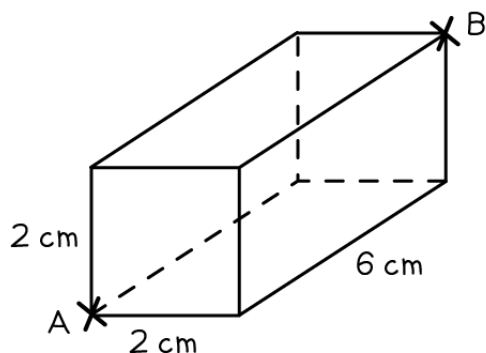
Copyright © Save My Exams. All Rights Reserved

- However, all 3D situations can be broken into **two 2D problems**
 - Form two right-angle triangles

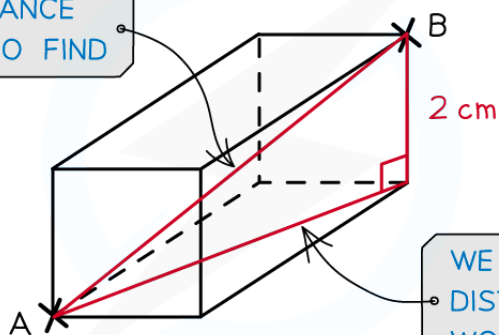


Your notes

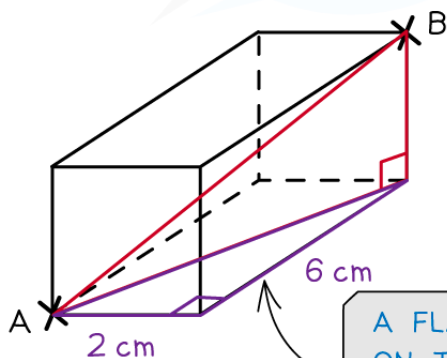
e.g. FIND THE DISTANCE BETWEEN A AND B.



THIS IS THE DISTANCE
WE ARE TRYING TO FIND



WE DON'T KNOW THIS
DISTANCE BUT CAN
WORK IT OUT

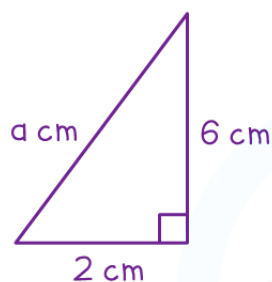


A FLAT RIGHT-ANGLED TRIANGLE
ON THE BASE ALLOWS US TO
FIND THE MISSING LENGTH OF
THE REAL TRIANGLE

Copyright © Save My Exams. All Rights Reserved

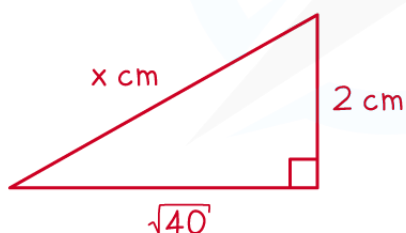


Your notes



NO NEED TO SQUARE ROOT - IT WILL GET SQUARED AGAIN SOON, ANYWAY

$$a^2 = 2^2 + 6^2 = 4 + 36 = 40$$



$$x^2 = (\sqrt{40})^2 + 2^2$$

$$x^2 = 40 + 4$$

$$x^2 = 44$$

$$x = 6.63 \quad (2 \text{ DP})$$

Copyright © Save My Exams. All Rights Reserved



Examiner Tips and Tricks

- You are not given the 3D Pythagoras formula in the exam
- You can always split 3D problems into two 2D problems (which don't need this formula)

How do I use SOHCAHTOA in 3D?

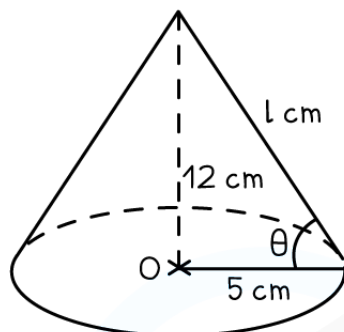
- Again, look for **right-angled triangles** to use with **SOHCAHTOA**
- You may need combinations of triangles that lead to the missing **side** or **angle**



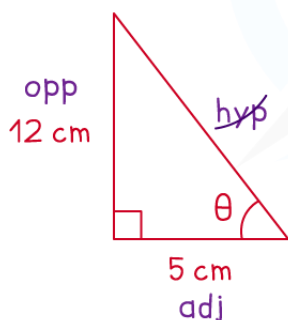
Your notes

SOHCAHTOA IN 3D

e.g. A CONE HAS BASE RADIUS 5 cm AND A PERPENDICULAR HEIGHT OF 12 cm.



FIND THE ANGLE θ , GIVING YOUR ANSWER TO ONE DECIMAL PLACE.



DRAW THE 2D TRIANGLE OUT TO MAKE IT EASIER TO SEE.

$$\text{TOA} - \tan \theta = \frac{\text{opp}}{\text{adj}}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{12}{5} = 2.4$$

$$\theta = \tan^{-1}(2.4)$$

$$\theta = 67.4 \text{ (1 DP)}$$

REMEMBER TO USE SHIFT WHEN FINDING ANGLES

Copyright © Save My Exams. All Rights Reserved

How do I find the angle between a line and a plane?

- To find the **angle** between a **line** and a **plane**:

- first **identity the plane** in the question
- then **draw** on the **line** in the question (if it is not already drawn on)
- form a **right-angled triangle** between the line and plane
 - The **height** of the triangle must be **perpendicular to the plane**
 - then use **SOHCAHTOA**
- The angle between a line and a plane is sometimes **not obvious**
 - If unsure, put a different point on the same line and see if it helps
 - Try to create a right-angled triangle
 - Some people like to imagine the line as a fishing rod, from which they lower the hook (and fishing line) vertically down to the plane!
 - Others like to imagine they are sitting inside the 3D object, looking around it (like corners of a room)

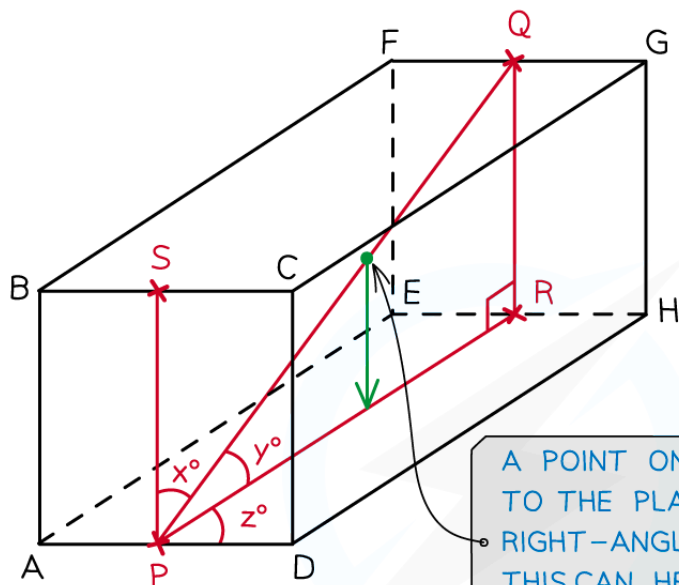


Your notes



Your notes

ANGLE BETWEEN A LINE AND A PLANE



A POINT ON THE LINE CONNECTED TO THE PLANE CREATES A RIGHT-ANGLED TRIANGLE AND THIS CAN HELP IDENTIFY THE ANGLE REQUIRED*

x° IS THE ANGLE BETWEEN THE LINE PQ AND THE PLANE ABCD (LINE PS)

y° IS THE ANGLE BETWEEN THE LINE PQ AND THE PLANE AEHD (LINE PR)

z° IS THE ANGLE BETWEEN THE LINE PR AND THE LINE AD

Copyright © Save My Exams. All Rights Reserved

How do I apply 3D Pythagoras and trigonometry to more complicated problems?

- Always split up a complicated problem into **2D right-angled triangles**
 - Some questions may require more than one 2D right-angled triangle

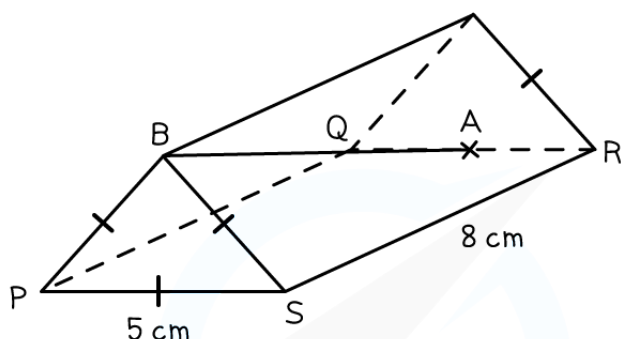
- Some 2D triangles on the diagram are still drawn in 3D
 - It helps to **redraw** these 2D triangles **flat on the page** (not at angles)
 - You can then spot any uses of Pythagoras' theorem and SOHCAHTOA



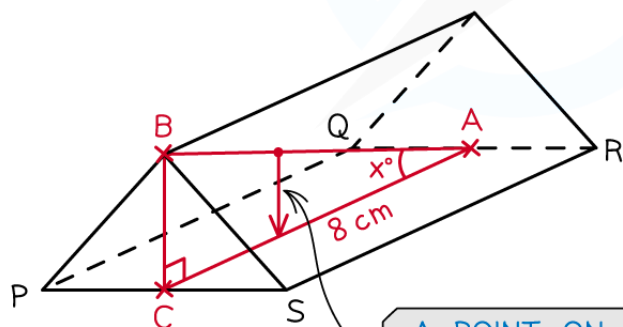
Your notes

PROBLEM SOLVING

e.g. IN THE TRIANGULAR PRISM BELOW, POINT A IS THE MID-POINT OF QR AND B IS A VERTEX OF THE FACE PBS WHICH IS AN EQUILATERAL TRIANGLE.



FIND THE ANGLE BETWEEN THE LINE AB AND THE PLANE PQRS.



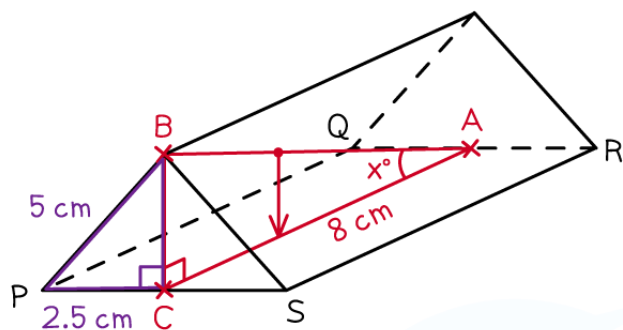
DRAW IN A RIGHT ANGLED TRIANGLE AND DETERMINE ANY KNOWN DISTANCES, AND THE ANGLE BEING ASKED FOR.

A POINT ON THE LINE AB CONNECTED TO THE PLANE PQRS CREATES A RIGHT ANGLED TRIANGLE

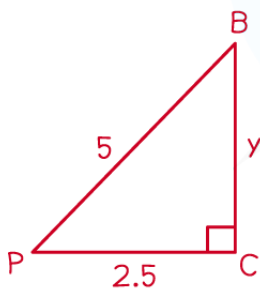
ONLY ONE SIDE OF TRIANGLE ABC IS KNOWN BUT BC CAN BE CALCULATED BY PYTHAGORAS' THEOREM.



Your notes



PB = 5 cm AS TRIANGLE PBS IS EQUILATERAL
PC = 2.5 cm AS A IS MIDPOINT OF QR,
SO C IS MIDPOINT OF PS.



DRAW THE 2D TRIANGLES
SEPARATELY TO HELP KEEP
THINGS CLEAR

$$y^2 = 5^2 - 2.5^2$$

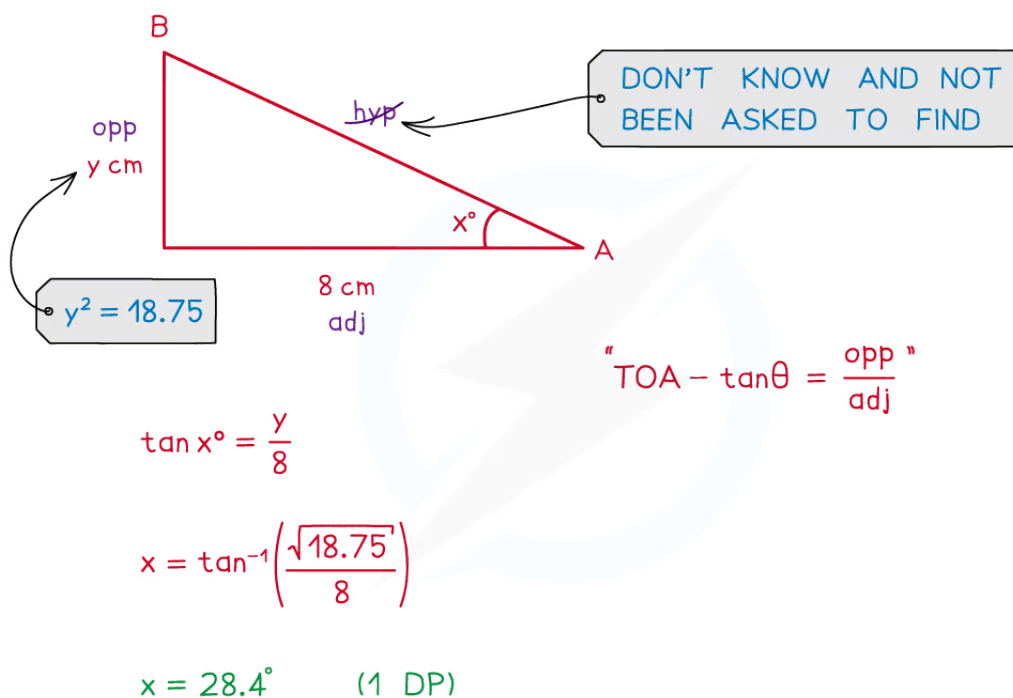
$$y^2 = 18.75$$

LEAVE AS y^2 FOR NOW TO
AVOID ROUNDING

Copyright © Save My Exams. All Rights Reserved



Your notes



$\tan x^\circ = \frac{y}{8}$
 $x = \tan^{-1}\left(\frac{\sqrt{18.75}}{8}\right)$
 $x = 28.4^\circ \quad (1 \text{ DP})$

TOA – $\tan \theta = \frac{\text{opp}}{\text{adj}}$



Examiner Tips and Tricks

- If you are stuck in the exam with a complicated 3D diagram, it is always better to just start finding any lengths and angles in the shape, as:
 - these may end up being useful
 - you may score more marks than if you had left the question blank



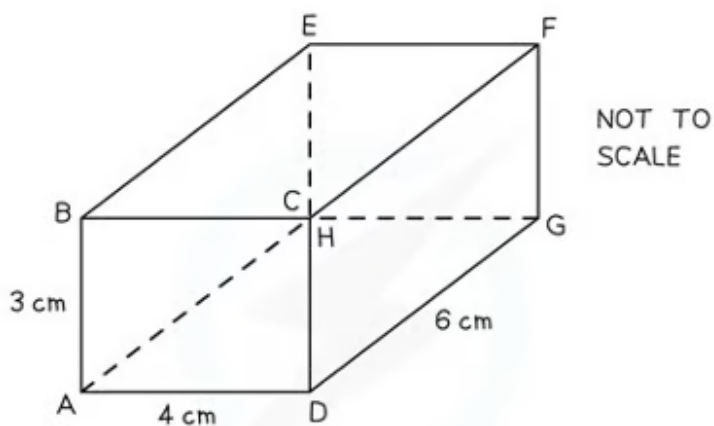
Worked Example

A pencil is being put into a cuboid shaped box.

The box has dimensions 3 cm by 4 cm by 6 cm.



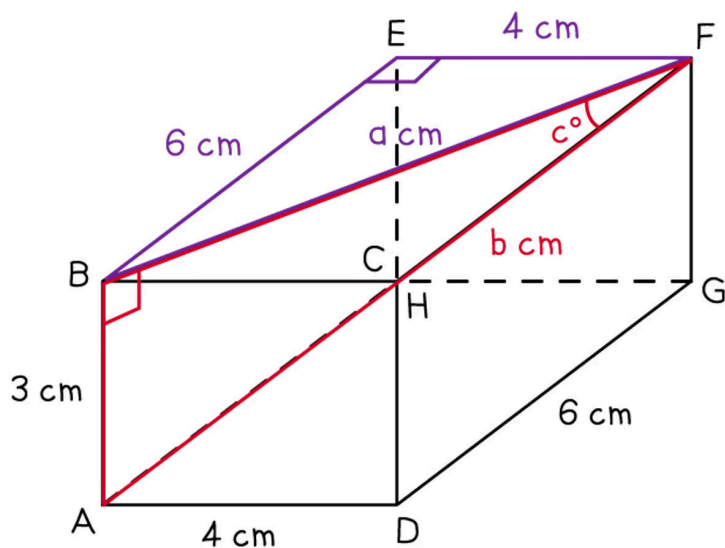
Your notes



(a) Find the length of the longest pencil that can fit inside the box.

The longest possible pencil will fit between diagonally opposite vertices, e.g. AF

Form a 2D right-angled triangle, such as triangle ABF



Method 1

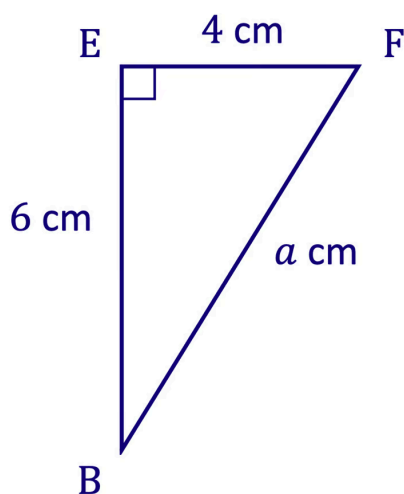


Your notes

To find the length AF, there are a few different options

One option is to find length BF (from triangle BEF) then AF (from triangle ABF)

Draw triangle BEF flat and use Pythagoras' theorem to find BF



$$a^2 = 4^2 + 6^2$$

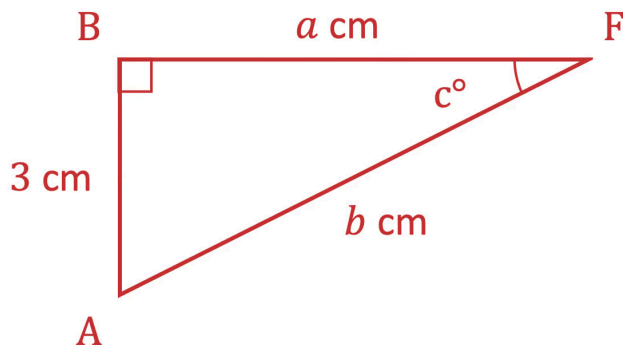
$$a^2 = 16 + 36$$

$$a^2 = 52$$

Draw triangle ABF flat and use Pythagoras' theorem to calculate AF



Your notes



$$b^2 = 3^2 + a^2$$

$$b^2 = 9 + 52$$

$$b^2 = 61$$

$$b = \sqrt{61} = 7.81024...$$

The longest pencil that can fit inside the box is 7.81 cm (to 3 s.f.)

Method 2

Apply the **3D version** of Pythagoras' theorem: $d^2 = x^2 + y^2 + z^2$

The distance in the x direction is 4 cm

The distance in the y direction is 6 cm

The distance in the z direction is 3 cm

$$d^2 = 4^2 + 6^2 + 3^2$$

$$d = \sqrt{4^2 + 6^2 + 3^2}$$

$$d = \sqrt{61} = 7.81024...$$

The longest pencil that can fit inside the box is 7.81 cm (to 3 s.f.)

(b) Find the angle that the pencil would make with the plane BEFC.

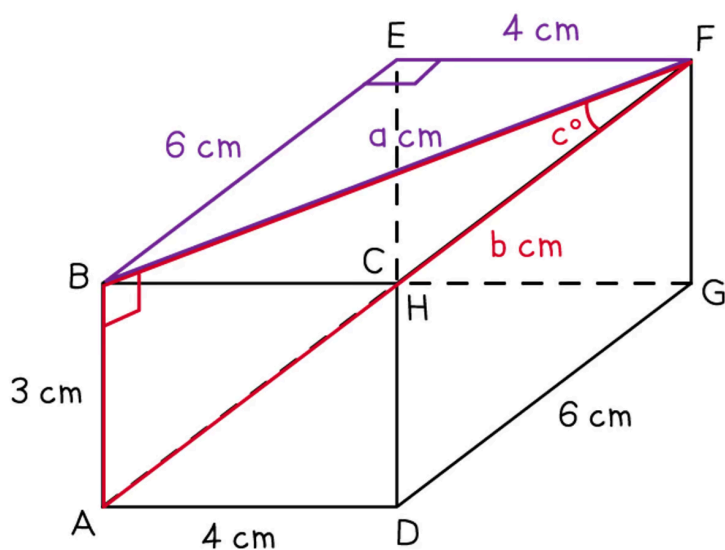
The plane BEFC is the horizontal top surface of the box

To see the angle between AF and the plane, form a triangle with side AB (which is a height perpendicular to the plane)

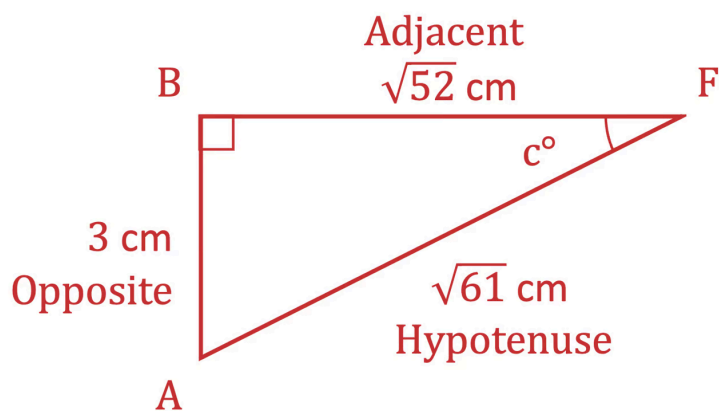


Your notes

The angle needed is marked c



Triangle ABF is the 2D triangle needed, but it is currently drawn in 3D
Draw triangle ABF flat on the paper and write on the lengths from part (a)





Your notes

This is now ready for SOHCAHTOA

We know all three sides so could use any trig ratio, for example $\tan \theta = \frac{O}{A}$

Use this to find angle c

$$\begin{aligned}\tan c &= \frac{3}{\sqrt{52}} \\ c &= \tan^{-1}\left(\frac{3}{\sqrt{52}}\right) \\ c &= 22.58853\dots\end{aligned}$$

The angle between plane BEFC and the pencil is 22.6° (to 1 d.p.)